# Harrison Felstead

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## PROFESSIONAL SUMMARY

Final-year Mechatronics (Honours) and Computer Science student at Monash University (completing June 2027), focused on AI development. Built a computer vision and A* pathfinding system for an autonomous robot, an unsupervised learning pipeline that turns foot pressure data into 3D-printed insoles, and a data logging app heading to a school pilot in 2027. Combines AI and software skills with PLC, embedded and CAD experience. Seeking internship and graduate roles in AI development with mentorship.

## AREAS OF EXPERTISE

- **AI development:** neural networks (PyTorch), unsupervised learning, computer vision (YOLO)
- **PLC control systems:** ladder logic and HMI (CODESYS), ISA-95, SCADA concepts
- **Hardware and software integration:** STM32, Arduino, Raspberry Pi
- **Testing, troubleshooting and verification** of robots and microprocessor systems
- **App development:** PostgreSQL, Expo, React, Tailwind
- **Data organisation**, process streamlining and technical documentation
- **CAD and additive manufacturing:** SolidWorks (CSWA), TPU 3D printing
- **Team leadership** and project delivery in teams of 3 to 5

## PROFESSIONAL EXPERIENCE

### ENGINEERING & TECHNICAL PROJECTS

**Final Year Project: Custom Orthopaedic Sole Creation using Unsupervised Learning** *2026 – Present*

*Monash University · Team of 3 (only Mechatronics student)*

- Developed an end-to-end workflow from raw plantar pressure data to a 3D-printable customised sole, using unsupervised learning (K-means) to group regions of similar pressure.
- Printed 5 TPU prototypes and validated them on a Pedar in-shoe pressure system, measuring pressure change across the gait cycle.
- Corrected incorrect material allocation on the pressure map by adding parameters and methods that follow accepted orthotic design practice, giving a significant improvement.
- With no existing quantitative benchmark, defined our own comparison measures; results indicate improvement without claiming a complete definition.
- Managed scope by cutting feature additions to focus on pressure reduction. Applied technical documentation, test planning and iteration from earlier mistakes.

**Fruit-Finding Robot: Pathfinding and Vision System** *Sem 2 2025*

*Monash University · Team of 4*

- Developed A* pathfinding and a YOLO vision system in Python so a robot could identify fruit and navigate around obstacles using only what it could see.
- Programmed and implemented the vision system and optimised parameters during testing.
- Robot collected all fruit within the time limit with no obstacle collisions.

**NCCD Mobile App and Website** *Early 2026 – Present*

*Client project (outside university)*

- Built a data logging system that removes roadblocks for teachers and school staff collecting the data used to support school funding, using PostgreSQL, Expo, React and Tailwind.
- Learned the fundamentals of regulated data storage, UI/UX design, and working with clients to deliver a product.
- Prototypes are performing well; a school pilot is scheduled for Term 1 2027.

**Carpark Carousel PLC System** *Sem 2 2026*

*Monash University · Team of 5*

- Designed a multi-PLC control system for a carpark carousel using CODESYS (ladder logic and HMI) and built a small prototype for demonstration, applying ISA-95 and SCADA concepts.
- Designed the safety PLC and the communication between PLCs, delivering dynamic fault detection and working multi-PLC communication.

**Research & Development Team Member** *Apr 2026 – Sep 2026*

*Monash BrewLab, Monash University*

- Researched new recipes and adapted them to our lab conditions and equipment, then produced them.
- Trained most new members to use the space and built their confidence with the equipment.

### LEADERSHIP

**Brewkeepers Lead** *2026 – Present*

*Monash BrewLab, Monash University · Sub-team of 3 to 5*

- Lead the sub-team responsible for lab upkeep, logistics and administration, keeping operations organised so the wider team can run its work.

**Education Officer** *Sep 2024 – Sep 2025*

*Monash Engineering Students Society (MESS), Monash University*

- Designed and wrote the "Ed Guide", a first-year resource adopted by the Engineering faculty.
- Mentored first-year Engineering students.

**Vice President** *Sep 2024 – Present*

*German Liechtenstein Austrian Swiss Society (GLASS), Monash University*

- Oversee committee operations, guide decision-making and plan the society's annual program, including cross-cultural events.

**Wellbeing & Equity Officer** *Sep 2023 – Sep 2024*

*Monash Engineering Students Society (MESS), Monash University*

- Designed and launched mental health awareness and inclusivity programs across the Engineering faculty.

### OTHER EMPLOYMENT

**OSHC Educator** *Summer 2025 – 2026*

*Holy Cross Primary School, Cairns, QLD*

- Supervised and guided primary school students in structured activities, building the communication and patience needed for client-facing work.

## EDUCATION AND QUALIFICATIONS

**Bachelor of Engineering (Honours) (Mechatronics) / Bachelor of Science (Computer Science)** *2022 – Present*

*Monash University, Melbourne, VIC · Expected completion June 2027*

- Coursework: neural networks and machine learning (PyTorch), control systems, PLC systems, signal processing, embedded systems (C/C++), robotics, CAD.
- Safety training: chemistry units covering risk assessments, PPE, and safe chemical handling and disposal.

**Queensland Certificate of Education** *2018 – 2021*

*St Augustine's Marist College, Cairns, QLD · ATAR: 97.80*

- **Certified SOLIDWORKS Associate (CSWA)**
- **Google Project Management Certificate**, Coursera, Dec 2023
- **Google IT Support Certificate**, Coursera, Dec 2023
- **Working with Children Check** held

## CONTINUING PROFESSIONAL DEVELOPMENT

- Anthropic: AI Fluency (Framework and Foundations; for Builders)
- Anthropic: Claude Code 101 and Claude Code in Action
- Anthropic: Introduction to Agent Skills and Subagents
- Anthropic: Model Context Protocol (Introduction and Advanced Topics); Building with Claude API

## SKILLS

**Programming:** Python, C++, C, Assembly, MATLAB
**AI and Data:** PyTorch, YOLO, K-means, Jupyter
**Web and App:** React, Expo, Tailwind, PostgreSQL
**Hardware and CAD:** CODESYS, STM32, Arduino, Raspberry Pi, SolidWorks
**Tools:** Git, Linux, Microsoft Office Suite, Agile / Scrum
**Languages:** English (Native), German (A2)

## INTERESTS

Surfing, hiking, and building quality-of-life apps.

## REFERENCES

*Available upon request.*